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Crop classification based on phenology information by using time series of optical and synthetic-aperture radar images

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Highlights

- Crop classification map by combination time series of Optical and SAR data.
- Extracting key phonological parameters of agriculture products by TIMESAT model.
- Evaluating and choose the best vegetation index and optical images for Classification.
- Identifying five major crops with overall accuracy of 89% using the support vector machine algorithm.

Abstract

Crop classification provides essential information for ensuring global food security, allowing early crop monitoring practices and water irrigation management. Despite of high importance of satellite imagery in identifying agricultural products, a significant number of optical images could become inaccessible due to cloud cover, which makes mapping agricultural products difficult. Therefore, the provision of up-to-date data integration methods and the availability of time series of optical and radar images such as Sentinel-1 data have provided new opportunities for mapping agricultural products with high spatial and temporal resolution. In this study, the results of classification accuracy were investigated based on the phenology of agricultural products and using time series of Landsat 8, Sentinel-1 and the digital elevation model (DEM). To do so, the key phenological parameters of the products such as start of season, length of season and end of season were first extracted using time series of the NDVI vegetation index calculated from the optical images and the TIMESAT model. Then, based on vegetation phenology of crops using optical (Landsat-8) and radar (Sentinel-1) time series data along with Digital Elevation Model (DEM) to increase the accuracy of classification, we applied the support vector machine method based on machine learning. The performance of this model was investigated on a part of Miandoab catchment basin located in the catchment of Lake Urmia in the northwest of Iran. Also, five major agricultural products of the study area, such as alfalfa, wheat, sugar beet, apple and grape were identified with overall accuracy of 89% using the support vector machine algorithm. Classification results demonstrated that using a combination of different data (overall accuracy (OA)=89%, Kappa=0.78) were more accurate than the only single-sensor inputs (OA=77%, Kappa=0.64), and when differences between crop types were largest, classification performance increased throughout the season until July.

Introduction

In the future, more agricultural products will be needed to provide food (Foley et al., 2011). In addition, biofuel production and urban growth will increase the pressure on the agricultural sector (Godfray et al., 2010). All of these factors will have consequences for natural ecosystems (Green et al., 2005). Knowledge about types of agricultural products can help managers and farmers to apply appropriate management and executive policies to determine the type of cultivation specific to each region, estimate the potential for harvesting, manage agricultural lands in accordance with local needs and conditions, manage water resources (Demarez et al., 2019). A map of crop sequences can be produced by combining information from multiple years, thus providing an indicator of agricultural land use intensity. Both duration and diversity of crop sequences are directly related to landscape complexity (Tscharntke et al., 2021).

One of the effective tools for monitoring, studying, and determining types of agricultural products is satellite data (Van Tricht et al., 2018). Remote sensing techniques are widely used in all sectors including agriculture due to their ability to provide update data and high image analysis (Bégué et al., 2018). In Iran, estimation of the area under cultivation of various agricultural products is usually made through three methods: expert estimation, estimation through cataloging, and the use of remote sensing and GIS (Kenduiywo et al., 2018). Studies have shown that the accuracy of the first and second methods is very low and they have many errors(Ziaeian-Firoozabadi et al., 2009).Therefore, it is necessary to use more accurate, faster, and less expensive methods to promote various macro-planning, including agriculture (Nirbhay Bhuyar, 2020). Biophysical parameters, including fresh and dry weight and leaf area index can also be obtained using vegetation indices calculated from Landsat 8 OLI and Landsat 7 ETM+ (Ahmadian et al., 2016).

However, in spite of potential of optical remote sensing data in crop classification, there are some questions about where meteorological conditions can affect the behavior of the optical signal especially in cloud condition. Similarly, even though SAR remote sensing is a good choice to overcome a part of the meteorological conditions, its potential to gain information on the physical characteristics and structure of crop is still unknown for the identification and characterization in detail of cloud condition crop patterns (Carrasco et al., 2019).

Research has suggested that full-polarization radar data are needed to achieve high-quality results in mapping crop types, but other researchers have come to different conclusions based on their experience. Moreover, although the use of reflection coefficients has an overall accuracy (OA) less than 50% (Roychowdhury, 2020), more accurate classification is possible using the combination of haralic textures, the VV/VH polarization ratio, and the local mean with VV reflection coefficients (Inglada et al., 2016).

Combination of optical data (Sentinel-2 or Landsat 8) and SAR data, such as Sentinel-1 data could improve the identification and differentiation between crops type, whose detection is difficult using only optical images (Vuolo et al., 2018). By combining optical, SAR, and environmental data, overall accuracy was increased by 6%–10% compared to single-sensor approaches, where optical data performed better than SAR. The mapped areas corresponded well to agricultural statistics at a regional and national level, and their accuracy ranged between 78% and 80% (Blickensdörfer et al., 2022).

Agricultural products mapping in different climates and cultivation systems using optical time series are presented (Sonobe et al., 2018). However, optical images affected by cloud cover reduce the performance of these images in mapping agricultural products. Some studies have shown that the phenological properties of agriculture products obtained from optical sensors are also useful for estimating the area under crop (Zhang et al., 2017). Therefore, to achieve this goal, combining optical and radar time series data could be a smart method due to the regular revisit times and the independence of radar images from weather conditions. Further studies have examined the suitability of X- and C-band SAR data to determine the growth stages of rice (Yuzugullu et al., 2017).

Likewise, the multi-temporal classification method has been demonstrated in several studies using a combination of optics and radar time series data (Larrañaga and Álvarez-Mozos, 2016). Most methods used to classify agricultural products have used all available data during the growing season, regardless of vegetation phenology of agricultural products (Skriver, 2012). Although these studies have yielded significant results on the classification of agricultural products, a number of them have encountered difficulties in distinguishing and identifying some products such as cereals (Bargiel and Herrmann, 2011). Therefore, knowledge about vegetation phenology of crops can play a decisive role in improving the results of agricultural products classification (Bargiel, 2017).

Furthermore, examining the views related to the production of agricultural products maps, it was concluded that it is not possible to accurately classify them by selecting images related to specific dates and all available data should be used (Inglada et al., 2016). However, in a study, using the Electro Magnetics Institute Synthetic Aperture Radar (EMISAR) airborne system showing monthly polarimetric data at 20% error rate in some areas, they could use the images taken at the most appropriate and optimal time for mapping agricultural products (Skriver, 2012). In another study, using multi-temporal ERS-1 data, 12 crops were separated with 80% accuracy (Schotten et al., 1995).To achieve this goal, they needed to choose the best images from 14 available im-ages during the agricultural season. Moreover, in a similar study, only two Sentinel-1 images were used to identify some land uses(Balzter et al., 2015).

In addition to good quality remote sensing data, classification algorithms are necessary to produce accurate maps of agricultural products. Based on multi-temporal NDVI curves, the RF and SVM classifiers performed better than the SAM classifier in identifying crops (Ouzemou et al., 2018). For example, in one study, four methods of supervise classification such as support vector machine classification (SVMs), decision trees (DTs), neural network classification, and maximum likelihood classification (MLC) were analyzed on Landsat TM data. Results showed that support vector machine (SVMs) outperformed the other three algorithms (Huang et al., 2007).

This study aims to assess the potential of optical (Landsat-8) and radar (Sentinel-1) time series data along with DEM for crop classification based on vegetation phenology when optimal remote sensing data are not available due to atmospheric effects such as cloud and haze contamination. The case study considered in this article is located in the northwestern part of Iran. To classify agricultural products, the support vector machine method based on machine learning algorithm was applied. Finally, the results of the classification were validated using ground data collected in 2014. The main reason for selecting the year 2014 was the availability of ground truth data fo this year in order to validation of the results of classification.

The location of the study area has been shown in Fig. 1. Miandoab plain, with an area of 438,000 ha in the northwestern part of Iran, forms a large part of western Miandoab, which is one of the suitable areas for agricultural production downstream of Lake Urmia catchment due to its favorable climate. The geographical coordinates of the study area is from 45° 43′ 11″ to 46° 50′ 42″ east longitude and from 36° 40′ 49″ to 37° 44′ 2″ north latitude. This region has a cold and semi-arid cold climate according to Umberg classification and a cold semi-arid climate according to De Martonne classification. The annual rainfall in the study area is about 254 mm.

To evaluate and validate the results, ground truth data are of great importance along with satellite data. In this study, this data included information about the crop calendar, land harvested points, and cadastral map. In Fig. 2, a few of the points and cadastral data retrieved from Google Earth are shown with area photos taken during the field visit. In the study area, the Urmia Lake Restoration Program (ULRP) and Novinabyari website provided most of the ground truth data during the growth period.

The ground truth data collected in this study were related to the 2014–2015 crop years and no data were available for the subsequent years. Therefore, since the Sentinel-2A was launched in June 2015, firstly, these images did not cover the growing season of the products in the desired years. Secondly, it was not possible to validate the results without ground data. Table 1 shows the area of major agricultural products in the study area and the percentages of these lands in relation to the total agricultural lands in the Miandoab catchment area. The number of points used in the classification were 18500, to avoid training data imbalances that might negatively affect the SVM algorithm, 50% training samples and 50% validation samples were used for each land-use classification.

The Landsat8 satellite was successfully launched in February 2013. This satellite uses OLI as well as infrared thermal sensors. Also, Landsat 8 have ability to acquire images from the surface of the earth with a spatial resolution of 30 m and a temporal resolution of 16 days, as a georeference likewise other satellites of the Landsat series. Each image frame obtained from the Landsat 8 satellite can cover an area, which is 170 km north-south wide and 183 km east-west long (Zanter, 2016). In the present study, processed Landsat 8 images of Level 2A (atmospherically corrected surface reflectance by masking clouds, cloud shadows, water, and snow) were used. The study area was covered by one Landsat 8 image frame. All Landsat 8 images were resampled by Cubic Convolution to 10 m to be combined with radar images (Torres et al., 2012). According to the study time, radar images of Sentinel-1 (Interferometric Wide Swath mode) with a 12-day revisit time were used. Currently, the revisit time is in the range of 3–6 days depending on the combination of S1-A and S1–B platforms. In Sentinel-1, synthetic aperture radar (SAR) data has been acquired in the C-band for a dual-polarization system. There are three resolutions (10, 25, or 40m) and four bands (corresponding to scene polarization) available in Sentinel-1 data. These images were taken with a spatial resolution of 10 m with a constant incidence angle in VV and VH polarizations. All images were Level 1-GRD (Ground Range Detected) and radiometric and orthorectified corrections were applied on them using SNAP software. Then, From the two original bands, two new bands were calculated as (VV+VH) and (VH/VV). Lee 3 filter was also used to reduce the speckle noise. The acquisition date of Landsat8 and Sentinel-1A images has been shown in Table 2. It should be also noted that Iota2 software was used to fill temporal gap of cloud pixels (Inglada et al., 2015). The used gap-filling model relied on linear interpolation. Moreover, masked cloud pixels were estimated by linear interpolation using pixel information of the images taken of a date after and a date before when it was not cloudy. Therefore, a Landsat-8 image became available every 16 days. Since different agricultural products are grown at different elevation, therefore, the effect of SRTM Digital Elevation Model with a spatial resolution of 30 m on classification was evaluated.

Section snippets

Materials and methods

In Fig. 3, the flowchart of the general implementation process of the study is shown. In this study, it was tried to identify and classify the agricultural products of the study area by using time series of optical and radar images. This goal was achieved through several stages. Firstly, the images of the Landsat8 satellite images with a spatial resolution of 30m and the Sentinel-1 satellite images with a spatial resolution of 10m were downloaded and processed. Secondly, various vegetation …

Investigating the trend of changes in vegetation index

Using the coordinates of the existing land points, the trend of changes in agricultural products in the study area was separately investigated in the calculated vegetation indices. As mentioned earlier, the main agricultural products of the region included: apples, grapes, alfalfa, wheat, sugar beet, and various orchards. The trend of changes in orchard class and apple were close to each other and due to the fact that these crops are tree-type and have high greenery during the growing period, …

Conclusion

This study was conducted to improve the accuracy in classification results based on crops phenology obtained from combining the time series of optical and radar data with the digital elevation model. The results showed that combining Sentinel-1 and Landsat 8 data improved classification accuracy compared to the case performed with only optical or radar images. The use of radar data limits the use of temporal gap-filled models to generate interpolated data from other images for filling masked …

Ethical statement for solid state ionics

Hereby, I/Hossein Yousefi/consciously assure that for the manuscript/Crop classification based on phenology information by using time series of optical and Synthetic-aperture radar images for optimizing management of water resources/the following is fulfilled:

1) This material is the authors' own original work, which has not been previously published elsewhere. …

2) The paper is not currently being considered for publication elsewhere. …

3) The paper reflects the authors' own research and analysis in a …

…

Author statement

Fatemeh Kordi: conceived and designed the Conceptualization, methodology, software, validation visualization and writing the draft preparation.

Hossein Yousefi: Contributed to the writing, review and editing and supervision the paper. …

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. …

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